

Introduction:

Multifocal glioblastomas (mGBMs) present as multiple contrast-enhancing lesions that are either connected through white-matter or vascular tracts or appear as distinct, unconnected foci (multicentric). Clinically, mGBM patients have shorter survival and more limited surgical options than those with unifocal GBM (uGBM). MRI, particularly contrast-enhanced T1-weighted and T2-FLAIR sequences, remains the primary method for detecting these lesions. However, emerging evidence suggests that some radiologically unifocal GBMs harbor microscopic secondary foci that later merge or become apparent on follow-up imaging, indicating that many mGBMs evolve clonally from an initially unifocal tumor.

Methods:

We developed a deep-learning framework to identify imaging biomarkers predictive of multifocal recurrence. A 3D nnU-Net segmentation model was first trained on the curated unifocal GBM cohort from the multi-institutional BraTS 2017 dataset and then fine-tuned on Johns Hopkins Hospital (JHH) data containing both unifocal and multifocal GBMs. Using learned feature representations from the nnU-Net encoder, a downstream classifier was trained to predict whether a unifocal GBM at diagnosis would recur as unifocal or multifocal. Model interpretability was assessed by visualizing the nnU-Net feature kernels most influential for recurrence prediction.

Results:

Preliminary analysis indicates that the model captures texture and intensity patterns characteristic of multifocal morphology, with discriminative features localized to ?. These learned features correspond to regions implicated in early clonal migration and spatial diversification.

Conclusion

Our results demonstrate the feasibility of using deep-learning–derived imaging biomarkers to predict multifocal recurrence in GBM. This interpretable framework could enhance risk stratification, guide post-operative surveillance, and support personalized therapeutic planning for patients with high-grade glioma.